

Anupam Roy

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Collaborative Backend Node.js Developer with 4+ years of experience in designing scalable microservices and APIs using **GraphQL, TypeScript, and PostgreSQL**. Passionate about building reliable, high-performance backend systems that enhance team productivity, streamline **CI/CD** workflows, and deliver seamless customer experiences in **Agile environments**.

Professional Experience:

Oct 2021 - Present

Software Engineer | Tata Consultancy Services (Client: Avis Budget Group) | Kolkata, India

- **Led an 8-member offshore Development team** to architect high-performance GraphQL microservices using **NestJS** and **TypeScript**, achieving **sub-15ms response times** through clustering, load balancing, and Redis caching. Optimized **AWS ElastiCache** costs by designing and implementing a data **compression strategy** using **Gzip and LZ4**, reducing individual Redis key sizes by **up to 25x**.
- Engineered a **real-time data replication pipeline** using **Kafka**, reliably processing and streaming messages to a **DB2 database** for critical data synchronization.
- Orchestrated the **end-to-end deployment lifecycle for microservices** on **AWS** using **KubectI** and **Argo CD**, ensuring high availability through a robust **Concourse CI/CD** pipeline.
- Strengthened data security by implementing end-to-end encryption for sensitive PCI/PII data using the **Voltage SDK**, ensuring full compliance with industry standards.
- Enhanced **system observability** by building interactive **Grafana dashboards with Prometheus** to monitor key service metrics, including CPU **utilization** and request **counts**.
- Automated database schema management by implementing CUD services with **Flyway**, enabling version-controlled migrations in a **PostgreSQL** environment.

Projects Experience:

Society Management System (Full Stack) Tech Stack: *PHP, MySQL, JavaScript, HTML*.

- Architected a web application to streamline society operations with distinct admin/member modules, improving efficiency for billing, complaints, and visitor tracking.

URL Extractor from Images (OCR) Tech Stack: *Python, OCR Libraries*

- Developed a Python utility that leverages OCR to automatically detect, extract, and launch hyperlinks from image files, eliminating manual data entry.

Education:

Heritage Institute of Technology, Kolkata, India

Aug 2017 - Oct 2021

B.Tech in Computer Science & Engineering | DGPA: 7.60

Technical Skills:

- **Languages:** TypeScript, JavaScript, Python, C, Java.
- **Backend & APIs:** GraphQL, NestJS, REST APIs, TypeORM.
- **Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis, DB2, Oracle SQL.
- **Cloud & DevOps:** AWS, Docker, KubectI, CI/CD (Concourse, Argo CD), Git, Bitbucket, Flyway.
- **Tools & Platforms:** Kafka, Grafana, Prometheus, Jira, Postman, Snyk, Erwin Data Modeler, Voltage SDK, Gzip, LZ4 .

Certifications / Achievements:

- **Special Initiative Award (2025), Applause for Team Award (2024), Xcelerate Warrior (2023)** – TCS
- **WBUT Aptitude Test – 40th Rank | TCS Internal Codevita Competition – 114th AIR**
- **Generative AI Foundation Curriculum -TCS | Programming with Python & Python Data Structures - Coursera**
- **Campus Ambassador – BITS Pilani APOGEE (2021), IIIT Allahabad (2020)**